# Assignment 2

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*) Total number of calls per year

→ OS A : 1000 calls

→ OS B : 2000 calls.

Total OS A + OS B = 1000 + 2000 = 3000 calls

Number of customers with a serious problem

→ OS A : 100

OS B : 120

Total serious problems calls 100 + 120 = 220

1) has a serious problem ?

$$P(\text{serious}) = \frac{\text{Total serious problems call}}{\text{Total number of calls}} = \frac{220}{3000}$$

$\boxed{P(\text{serious}) \approx 0.07333}$

2) runs OS A ?

$$P(\text{OS A}) = \frac{\text{Number of OSA calls}}{\text{Total number of OSA calls}} = \frac{1000}{3000} = \frac{1}{3}$$

$\boxed{P(\text{OS A}) \approx 0.333}$

3) runs OS B ?

$$P(\text{OS B}) = \frac{\text{Number of OS B calls}}{\text{Total number of OS B calls.}} = \frac{2000}{3000} = \frac{1}{2}$$

$\boxed{P(\text{OS B}) \approx 0.6667}$

4) runs OS A & has a serious problem ?

$$P(\text{OS A \& serious}) = \frac{\text{Number of serious OSA}}{\text{Total number of calls}} = \frac{100}{3000}$$

$\boxed{P(\text{OSA \& serious}) \approx 0.0333}$

5) Runs OS B & has a serious problem?

$$P(\text{OS B \& serious}) = \frac{\text{Number of serious OS B}}{\text{Total number of calls.}} = \frac{120}{3000}$$

$$\boxed{P(\text{OS B \& serious}) = 0.04}$$

6) Has a serious problem, while we already know the customer runs OS A?

$$P(\text{serious} \mid \text{OS A}) = \frac{P(\text{OS A \& serious})}{P(\text{OS A})} = \frac{0.0333}{0.333}$$

$$\boxed{P(\text{Serious} \mid \text{OS A}) = 0.1}$$

7) has a serious problem, while we already know the customer runs OS B!

$$P(\text{serious} \mid \text{OS B}) = \frac{P(\text{OS B \& serious})}{P(\text{OS B})} = \frac{0.04}{0.6667}$$

$$\boxed{P(\text{serious} \mid \text{OS B}) = 0.06}$$

8) Runs OS A, while we already know the customer has a serious problem?

$$P(\text{OS A} \mid \text{serious}) = \frac{P(\text{OS A \& serious})}{P(\text{serious})} = \frac{0.0333}{0.0733}$$

$$\boxed{P(\text{OS A} \mid \text{serious}) \simeq 0.4545}$$

9) Runs OS B, while we already know the customer has a serious problem?

$$P(\text{OS B} \mid \text{serious}) = \frac{P(\text{OS B \& serious})}{P(\text{serious})} = \frac{0.04}{0.0733}$$

$$\boxed{P(\text{OS B} \mid \text{serious}) \simeq 0.5455}$$

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10) Does not have a serious problem, while we ~~already~~ know the custome runs OS A?

$P(\text{Not serious} | OS\ A) = 1 - P(\text{serious} | OS\ A)$

$= 1 - 0.1 = 0.9$

$\boxed{P(\text{Not serious} | OS\ A) = 0.9}$

11) Does not have a problem, while we already know the customer runs OS B?

$P(\text{Not serious} | OS\ B) = 1 - P(\text{serious} | OS\ B)$

$= 1 - 0.06$

$\boxed{P(\text{Not serious} | OS\ B) = 0.94}$

12) Runs OS A, while we already know the customer does not have a serious problem?

* Customers donot have serious problems:

Total of not serious $= 3000 - 220 = \boxed{2780}$

* Number of OS A customers who do not have serious problems is

OS A not serious $= 1000 - 100 = \boxed{900}$

So,

$P(OS\ A | \text{Not Serious}) = \frac{900}{2780} = 0.3237..$

$\boxed{P(OS\ A | \text{Not Serious}) \approx 0.3237}$

13) Runs OS B, while we already know the customer does not have serious problem?

* Number of OS B customers who do not have problems

OS B not serious $= 2000 - 120 = \boxed{1880}$

So,

$P(OS\ B | \text{Not serious}) = \frac{1880}{2780} \approx \boxed{0.6763}$

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3)

Messages_A = 10,000

Message_B = 20,000

distruption_A = 0.05

distruption_B = 0.01

Total_messages = Message_A + message_B.

1) The receiver R receives a message, what is probability that

→ The message is from A?

→ The message is from B?

→ The message is ok?

→ The message is disrupted?

solution

→ from A

$$P(A) = \frac{10,000}{30,000} = \frac{1}{3} = 0.333333333\ldots$$

$$\boxed{P(A) \approx 0.3333}$$

→ from B

$$P(B) = \frac{20,000}{30,000} = \frac{2}{3} = 0.666666666\ldots$$

$$\boxed{P(B) \approx 0.6667}$$

→ message is ok.

$$P(OK) = P(OK|A) \cdot P(A) + P(OK|B) + P(B)$$

$$= (1-0.05) \cdot 0.3333 + (1-0.01) + 0.6667$$

$$= 0.95\,(0.3333) + 0.99\,(0.6667)$$

$$P(OK) \approx 0.3167 + 0.6600$$

$$\boxed{P(OK) = 0.9767}$$

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* message is disrupted ?

$P(\text{Disrupted}) = 1 - P(\text{OK})$

$P(\text{Disrupted}) = 1 - 0.9767$

$\boxed{P(\text{Disrupted}) = 0.0233}$

2) Generally, what is probability that.

→ a message from A is disrupted ?
→ a message from A is ok ?
→ a message from B is disrupted ?
→ a message from B is ok ?
→ a disrupted message is from A ?
→ an ok message is from A ?
→ a disrupted message is from B ?
→ an ok message is from B ?

solution

→ $P(\text{Disrupted}|A) = \boxed{0.05}$

→ $P(\text{OK}|A) = 1 - P(\text{Disrupted}|A) = 1 - 0.5$

$\boxed{P(\text{OK}|A) = 0.95}$

→ $P(\text{Disrupted}|B) = \boxed{0.01}$

→ $P(\text{OK}|B) = 1 - P(\text{Disrupted}|B) = 1 - 0.01$

$\boxed{P(\text{OK}|B) = 0.99}$

→ $P(A|\text{Disrupted}) = \dfrac{P(\text{Disrupted}|A) \cdot P(A)}{P(\text{Disrupted})}$

$P(A|\text{Disrupted}) = \dfrac{0.05 \cdot 0.3333}{0.0233}$

$\boxed{P(A|\text{Disrupted}) \approx 0.7150}$

→ $P(A|\text{OK}) = \dfrac{P(\text{OK}|A) \cdot P(A)}{P(\text{OK})} = \dfrac{0.95 \cdot 0.3333}{0.9767}$

$\boxed{P(A|\text{OK}) \approx 0.3243}$

$\rightarrow$ $P(B|Disrupted) = \dfrac{P(Disrupted|B) \cdot P(B)}{P(Disrupted)} = \dfrac{0.01 * 0.6667}{0.0233}$

$\boxed{P(B|Disrupted) \approx 0.2850}$

$\rightarrow$ $P(B|ok) = \dfrac{P(OK|B) \cdot P(B)}{P(ok)} = \dfrac{0.99 * 0.6667}{0.9767}$

$\boxed{P(B|ok) \approx 0.6757}$

3. The receiver R receives 100 new message, how many messages in most probably disrupted.
$\rightarrow$ what is the probability that 0 or 1 or 2 or ... or 100 messages are disrupted ?
$\rightarrow$ what is the probability that 0 or max 1 or max 2 or max 100 messages are disrupted.

Solution

$P(Disrupted) = 0.0233$ the probability that a single message is disrupted

Number of trail $n = 100$.

$K$ is number of disrupted message

$P(Disrupted) \approx 0.0233$

$P(\text{Not Disrupted}) = 1 - P(\text{Disrupted}) = 0.9767$

$$P(X=K) = \binom{n}{K} \cdot (P(Disrupted))^K \cdot (1-P(Disrupted))^{n-k}$$

$$\binom{n}{K} = \frac{n!}{K!(n-K)!}$$

* $K=0$.

$$P(X=0) = \binom{100}{0} \cdot (0.0233)^0 \cdot (0.9767)^{100}$$

$$P(X=0) = 1 \cdot 1 \cdot (0.9767)^{100}$$

$\boxed{P(X=0) \approx 0.094}$

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* $k = 1$

$$P(X=1) = \binom{100}{1} \cdot (0.0233)^1 \cdot (0.9767)^{99}$$

$$P(X=1) = 100 * 0.0233\,(0.9767)^{99}.$$

$$\boxed{P(X=1) \approx 0.225}$$

* $k = 2$.

$$P(X=2) = \binom{100}{2} \cdot (0.0233)^2 \cdot (0.9767)^{98}.$$

$$P(X=2) = \frac{100 * 99}{2} \cdot (0.02333)^2 \cdot (0.9767)^{98}$$

$$\boxed{P(X=2) \approx 0.262}$$

→

$$P(X \le k) = \sum_{i=0}^{k} P(X=i)$$

$$P(X \le 0) = P(X=0) \approx \boxed{0.094}$$

$$P(X \le 1) = P(X=0) + P(X=1) \approx 0.094 + 0.225$$

$$\boxed{0.319}$$

$$P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$$

$$\approx 0.094 + 0.225 + 0.262 = \boxed{0.581}$$

* $k = 0$.

$$P(X=0) \approx 0.09465, \quad P(X \le 0) \approx \boxed{0.09465}$$

$$P(X=1) \approx 0.22579, \quad P(X \le 1) \approx \boxed{0.32044}$$

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Total messages : 1000

Messages received with disruption (NOK) = $\boxed{300}$

Messages received without disruption (OK)

1000 - 300 = $\boxed{700}$

Detection results

Correctly detected disrupted messages (True Positives, TP) : $\boxed{200}$

Mistakenly detected 50 OK messages as disrupted (False Positives, FP) : $\boxed{50.}$

So,

False Negatives (FN) : Disrupted messages that were not detected

300 - 200 = $\boxed{100}$

True Negatives (TN) : OK messages correctly detected as OK    700 - 50 = $\boxed{650}$

→ Sensitivity (S)

$$S = \frac{TP}{TP+FN} = \frac{200}{200+100} = \frac{200}{300} = 0.6666.6\ldots$$

$\boxed{\text{Sensitivity} \approx 0.6667}$

$$\rightarrow \text{Specificity} = \frac{TN}{TN+FP} = \frac{650}{650+50} = \frac{650}{700}$$

$\boxed{\text{Specificity} \approx 0.929}$

True Positive Rate (TPR) = Sensitivity = $\boxed{0.6667}$

False Positive Rate (FPR) = 1 - Specificity = 1 - 0.929

= $\boxed{0.071.}$

Point on ROC curve (0.071, 0.667)